CAR RENTAL SYSTEM DATABASE

Physical Data Model Documentation

Option 4 — Car Rental System | 3NF | 12 Tables

CONTENTS

1 BUSINESS DESCRIPTION

1.1 Business Background

Car rental companies manage large fleets of vehicles across multiple branches, serving thousands of customers. Tracking reservations, active rentals, vehicle conditions, payments, and maintenance activities across different locations is complex and error-prone without a centralized system.

A reliable database is needed to manage vehicles, customers, rental locations, reservations, return logs, vehicle condition reports, payment details, and maintenance tracking while supporting real-time availability queries.

1.2 Problems. Current Situation

Without a centralized database, the company faces:

- No unified tracking of reservations and active rentals across branches.
- Manual recording of vehicle condition at pickup and return — leading to disputes.
- No systematic maintenance scheduling, causing unexpected vehicle breakdowns.
- Payment records scattered across paper receipts or unlinked spreadsheets.
- Inability to check real-time vehicle availability across all locations.

1.3 Benefits of Implementing a Database. Project Vision

Implementing this relational database will:

- Centralize all rental, customer, vehicle, and branch data in one system.
- Automate availability tracking through vehicle status and reservation records.
- Maintain a full audit trail of vehicle condition at every pickup and return.
- Support multiple payment types and track payment status per rental.
- Schedule and track all vehicle maintenance to reduce downtime.
- Allow staff at any branch to access up-to-date rental and customer information.

2 MODEL DESCRIPTION

2.1 Definitions & Acronyms

Term	Definition
PK	Primary Key — unique identifier for each record in a table.
FK	Foreign Key — a reference to a primary key in another table.
Customer	A registered person who rents vehicles.
Rental	An active or completed vehicle rental agreement.
Reservation	A pre-booking made by a customer before a rental starts.
RentalLocation	A branch where vehicles can be picked up or returned.
Vehicle	A car in the company fleet available for rental.
VehicleType	Category of vehicle (Sedan, SUV, Luxury) with base daily rate.
Payment	A financial transaction linked to a rental.
VehicleConditionReport	Inspection record of a vehicle at pickup or return.
Maintenance	A scheduled or completed service/repair for a vehicle.
AdditionalServices	Optional add-ons (GPS, insurance, child seat) added to a rental.
RentalServices	Bridge table linking rentals to additional services (Many-to-Many).

2.2 Logical Scheme

The diagram below shows all 12 tables, their fields, and relationships (3NF):

Relationships summary:

- Customers → Rentals (1:N) — one customer can have many rentals
- Customers → Reservations (1:N) — one customer can make many reservations
- VehicleTypes → Vehicles (1:N) — one type covers many vehicles
- Vehicles → Rentals (1:N) — one vehicle can be rented many times
- Vehicles → Reservations (1:N) — one vehicle can be reserved many times
- Vehicles → Maintenance (1:N) — one vehicle has many maintenance records
- Vehicles → VehicleConditionReports (1:N) — one vehicle has many condition reports
- RentalLocations → Rentals (1:N) — pickup & return branches (2 FK)
- RentalLocations → Reservations (1:N) — pickup & return branches (2 FK)
- RentalLocations → Employees (1:N) — one branch employs many staff
- Reservations → Rentals (1:1 optional) — confirmed reservation converts to rental
- Rentals → Payments (1:N) — one rental can have multiple payments
- Rentals → VehicleConditionReports (1:N) — pickup + return reports per rental
- Rentals → RentalServices (1:N) — bridge for Many-to-Many
- AdditionalServices → RentalServices (1:N) — bridge for Many-to-Many
- Employees → VehicleConditionReports (1:N) — employee inspects vehicle

2.3 Objects

In any type of relationship, the first one will be a table that describes (one to many — one for the described table).

Table Name: Customers

Stores customer personal data. Every customer can make multiple rentals and reservations.

Field Name	Field Description	Data Type / Constraints
customer_id	Unique customer ID	INT, PK, AUTO_INCREMENT
first_name	First name	VARCHAR(50), NOT NULL
last_name	Last name	VARCHAR(50), NOT NULL
passport_number	Passport number	VARCHAR(30), UNIQUE, NOT NULL
driving_license	Driver's license number	VARCHAR(30), UNIQUE, NOT NULL
phone	Phone number	VARCHAR(20), NOT NULL
email	Email address	VARCHAR(100), UNIQUE
address	Home address	VARCHAR(255)

Relationships

Rentals (1:N), Reservations (1:N) — one customer can have many rentals and reservations.

Example with data

customer_id	first_name	last_name	passport_number	driving_license	phone
1	Aibek	Seitkali	AB1234567	DL9876543	+7 701 111 2233
2	Dana	Muratova	AB7654321	DL1234567	+7 702 222 3344

Table Name: RentalLocations

Stores rental branch data — pickup and return points for vehicles.

Field Name	Field Description	Data Type / Constraints
location_id	Branch ID	INT, PK, AUTO_INCREMENT
location_name	Branch name (e.g. Aktau Airport)	VARCHAR(100), NOT NULL
address	Street address	VARCHAR(255), NOT NULL
city	City	VARCHAR(50), NOT NULL
phone	Phone number	VARCHAR(20)

Relationships

Rentals (1:N), Reservations (1:N), Employees (1:N) — one branch handles many rentals, reservations, and employees.

Example with data

location_id	location_name	address	city	phone

1	Aktau Airport	Terminal 1, Aktau Airport	Aktau	+7 729 222 3344
2	Aktau Center	Nurzhol Ave 12	Aktau	+7 729 333 4455

Table Name: VehicleTypes

Reference table for vehicle categories with a base daily rate.

Field Name	Field Description	Data Type / Constraints
type_id	Type ID	INT, PK, AUTO_INCREMENT
type_name	Type name (Sedan, SUV, Luxury, etc.)	VARCHAR(50), NOT NULL
daily_rate	Base daily rental price	DECIMAL(10,2), NOT NULL

Relationships

Vehicles (1:N) — one vehicle type covers many vehicles.

Example with data

type_id	type_name	daily_rate
1	SUV	15000.00
2	Sedan	9000.00
3	Luxury	30000.00

Table Name: Vehicles

Stores the vehicle fleet. Each vehicle belongs to one type and can appear in many rentals, reservations, maintenance records, and condition reports.

Field Name	Field Description	Data Type / Constraints
vehicle_id	Vehicle ID	INT, PK, AUTO_INCREMENT
license_plate	License plate number	VARCHAR(20), UNIQUE, NOT NULL
make	Brand (Toyota, BMW, etc.)	VARCHAR(50), NOT NULL
model	Model name	VARCHAR(50), NOT NULL
year	Year of manufacture	YEAR, NOT NULL
type_id	Vehicle type FK	INT, FK → VehicleTypes, NOT NULL
status	Current availability	ENUM('Available','Rented','Maintenance','Reserved'), DEFAULT 'Available'
current_mileage	Odometer reading (km)	INT, DEFAULT 0

Relationships

VehicleTypes (N:1), Rentals (1:N), Reservations (1:N), Maintenance (1:N), VehicleConditionReports (1:N).

Example with data

vehicle_id	license_plate	make	model	year	type_id	status	current_mileage
1	160 AKT 01	Toyota	Land Cruiser	2022	1	Available	12000
2	161 AKT 02	BMW	5 Series	2023	3	Rented	5400

Table Name: Employees

Stores employee data. Each employee belongs to one branch and can inspect vehicles.

Field Name	Field Description	Data Type / Constraints
employee_id	Employee ID	INT, PK, AUTO_INCREMENT
first_name	First name	VARCHAR(50), NOT NULL
last_name	Last name	VARCHAR(50), NOT NULL
position	Job title	VARCHAR(50)
location_id	Branch FK	INT, FK → RentalLocations, NOT NULL

Relationships

RentalLocations (N:1), VehicleConditionReports (1:N).

Example with data

employee_id	first_name	last_name	position	location_id
1	Daniyar	Bekov	Manager	1
2	Asel	Nurova	Inspector	1

Table Name: Reservations

Stores pre-rental bookings. A confirmed reservation converts to an active Rental record.

Field Name	Field Description	Data Type / Constraints
reservation_id	Reservation ID	INT, PK, AUTO_INCREMENT
customer_id	Customer FK	INT, FK → Customers, NOT NULL
vehicle_id	Vehicle FK	INT, FK → Vehicles, NOT NULL
pickup_location_id	Pickup branch FK	INT, FK → RentalLocations, NOT NULL
return_location_id	Return branch FK	INT, FK → RentalLocations, NOT NULL
planned_start	Planned pickup date/time	DATETIME, NOT NULL
planned_end	Planned return date/time	DATETIME, NOT NULL
status	Reservation status	ENUM('Pending','Confirmed','Cancelled','Converted'), DEFAULT 'Pending'
created_at	Creation timestamp	DATETIME, DEFAULT CURRENT_TIMESTAMP

Relationships

Customers (N:1), Vehicles (N:1), RentalLocations (N:1) x2 (pickup & return), Rentals (1:1 optional).

Example with data

reservation_id	customer_id	vehicle_id	pickup_location_id	planned_start	planned_end	status
1	1	1	1	2025-06-01 08:00	2025-06-05 08:00	Confirmed
2	2	2	2	2025-06-10 09:00	2025-06-12 09:00	Pending

Table Name: Rentals

Central table of active and completed rentals. Links customer, vehicle, two branches, and optionally a prior reservation.

Field Name	Field Description	Data Type / Constraints
rental_id	Rental ID	INT, PK, AUTO_INCREMENT
customer_id	Customer FK	INT, FK → Customers, NOT NULL
vehicle_id	Vehicle FK	INT, FK → Vehicles, NOT NULL
pickup_location_id	Pickup branch FK	INT, FK → RentalLocations, NOT NULL
return_location_id	Return branch FK	INT, FK → RentalLocations, NOT NULL
reservation_id	Linked reservation FK (optional)	INT, FK → Reservations, NULL
rental_start	Rental start date/time	DATETIME, NOT NULL
rental_end	Planned return date/time	DATETIME, NOT NULL
actual_return	Actual return date/time	DATETIME
total_amount	Total amount due	DECIMAL(12,2)
status	Rental status	ENUM('Active','Completed','Cancelled','Overdue'), DEFAULT 'Active'

Relationships

Customers (N:1), Vehicles (N:1), RentalLocations (N:1) x2, Reservations (N:1 optional), Payments (1:N), VehicleConditionReports (1:N), RentalServices (1:N).

Example with data

rental_id	customer_id	vehicle_id	pickup_location_id	rental_start	rental_end	actual_return	status
1	1	1	1	2025-06-01 09:00	2025-06-05 09:00	2025-06-05 08:45	Completed
2	2	2	2	2025-06-10 09:00	2025-06-12 09:00		Active

Table Name: AdditionalServices

Reference table of optional add-ons (GPS, child seat, full insurance, etc.).

Field Name	Field Description	Data Type / Constraints
service_id	Service ID	INT, PK, AUTO_INCREMENT

Field Name	Field Description	Data Type / Constraints
service_name	Service name	VARCHAR(100), NOT NULL
daily_price	Daily price for the service	DECIMAL(8,2), NOT NULL

Relationships

RentalServices (1:N) — participates in Many-to-Many with Rentals via RentalServices bridge.

Example with data

service_id	service_name	daily_price
1	GPS Navigator	1500.00
2	Child Seat	800.00
3	Full Insurance	5000.00

Table Name: RentalServices

Bridge table implementing Many-to-Many between Rentals and AdditionalServices. One rental can include several services; one service can appear in many rentals.

Field Name	Field Description	Data Type / Constraints
rental_service_id	Record ID	INT, PK, AUTO_INCREMENT
rental_id	Rental FK	INT, FK → Rentals, NOT NULL
service_id	Service FK	INT, FK → AdditionalServices, NOT NULL
quantity	Quantity of units	INT, DEFAULT 1, NOT NULL
total_price	Total price for this line	DECIMAL(10,2), NOT NULL

Relationships

Rentals (N:1), AdditionalServices (N:1).

Example with data

rental_service_id	rental_id	service_id	quantity	total_price
1	1	1	1	6000.00
2	1	3	1	20000.00

Table Name: Payments

Tracks all payments linked to a rental. One rental can have multiple payments (deposit, full payment, penalty).

Field Name	Field Description	Data Type / Constraints
payment_id	Payment ID	INT, PK, AUTO_INCREMENT
rental_id	Rental FK	INT, FK → Rentals, NOT NULL
amount	Payment amount	DECIMAL(12,2), NOT NULL

Field Name	Field Description	Data Type / Constraints
payment_date	Date and time of payment	DATETIME, NOT NULL
payment_method	Payment method	ENUM('Cash','Card','Bank Transfer','Online'), NOT NULL
payment_type	Type of payment	ENUM('Deposit','Full','Partial','Penalty'), NOT NULL
status	Payment status	ENUM('Pending','Completed','Refunded'), DEFAULT 'Pending'
notes	Additional notes	VARCHAR(255)

Relationships

Rentals (N:1) — one rental can have many payment records.

Example with data

payment_id	rental_id	amount	payment_date	payment_method	payment_type	status
1	1	60000.00	2025-06-05 09:00	Card	Full	Completed
2	2	15000.00	2025-06-10 09:00	Cash	Deposit	Completed

Table Name: VehicleConditionReports

Records vehicle inspection at pickup and return. Captures mileage, fuel level, and damage notes.

Field Name	Field Description	Data Type / Constraints
report_id	Report ID	INT, PK, AUTO_INCREMENT
rental_id	Rental FK	INT, FK → Rentals, NOT NULL
vehicle_id	Vehicle FK	INT, FK → Vehicles, NOT NULL
employee_id	Inspector FK	INT, FK → Employees, NOT NULL
report_type	Inspection type	ENUM('Pickup','Return'), NOT NULL
mileage_at_report	Odometer at inspection (km)	INT, NOT NULL
fuel_level	Fuel level percentage	TINYINT, NOT NULL
condition_notes	Damage or condition notes	TEXT
reported_at	Inspection timestamp	DATETIME, DEFAULT CURRENT_TIMESTAMP

Relationships

Rentals (N:1), Vehicles (N:1), Employees (N:1) — each report links to one rental, one vehicle, and one inspector.

Example with data

report_id	rental_id	vehicle_id	employee_id	report_type	mileage_at_report	fuel_level	condition_notes
1	1	1	2	Pickup	12000	100	No damage
2	1	1	2	Return	12450	75	Small scratch on rear bumper

Table Name: Maintenance

Tracks all scheduled and completed maintenance for each vehicle. Supports availability management.

Field Name	Field Description	Data Type / Constraints
maintenance_id	Maintenance record ID	INT, PK, AUTO_INCREMENT
vehicle_id	Vehicle FK	INT, FK → Vehicles, NOT NULL
maintenance_type	Type of work	ENUM('Oil Change','Tire Change','Repair','Inspection','Other'), NOT NULL
description	Detailed work description	TEXT
cost	Maintenance cost	DECIMAL(10,2)
scheduled_date	Scheduled date	DATE, NOT NULL
completed_date	Actual completion date	DATE
status	Maintenance status	ENUM('Scheduled','In Progress','Completed','Cancelled'), DEFAULT 'Scheduled'
performed_by	Technician or company name	VARCHAR(100)

Relationships

Vehicles (N:1) — one vehicle can have many maintenance records.

Example with data

maintenance_id	vehicle_id	maintenance_type	cost	scheduled_date	completed_date	status
1	1	Oil Change	25000.00	2025-05-15	2025-05-15	Completed
2	2	Inspection	10000.00	2025-06-20		Scheduled

Assignment Requirements — Checklist

Requirement	Status & Notes
3NF (Third Normal Form)	✅ All 12 tables satisfy 3NF — no partial or transitive dependencies.
8–15 tables	✅ 12 tables total.
At least one Many-to-Many	✅ Rentals ↔ AdditionalServices via RentalServices bridge table.
All relationships shown on diagram	✅ 18 FK relationships shown with crow's foot notation.
Correct data types	✅ INT, VARCHAR, DECIMAL, DATETIME, DATE, TINYINT, TEXT, ENUM, YEAR used appropriately.
Keys and constraints (incl. bridge table)	✅ PK, FK, NOT NULL, UNIQUE, DEFAULT, AUTO_INCREMENT defined on all tables.
Clear table and column names	✅ Self-explanatory English names in snake_case convention.
Vehicles & availability	✅ Vehicles.status field + Maintenance + Reservations tables.
Reservations	✅ Reservations table — pre-rental booking with status tracking.
Return logs / Condition reports	✅ VehicleConditionReports — pickup & return inspection records.
Payment details	✅ Payments — multiple payments per rental, method, type, status.
Maintenance tracking	✅ Maintenance — scheduled & completed maintenance per vehicle.